



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0021



**Agricultural Machinery – Laser Leveler
LANDKING ALPHA XD-1000**

Test Report Number	2023 – 0021
Date of Issuance	February 21, 2023
Valid Until	February 21, 2028
Agricultural Machinery	Laser Leveler
Type	Trailer type
Brand	LANDKING
Model	ALPHA XD-1000
Test Requested By	FORD TRACTOR PHILS., INC. 1384, JOSE ABAD SANTOS, TONDO, MANILA

SUMMARY

Item	Manufacturer's Specification	AMTEC Test
Actual Field Capacity, ha/h	ND ^a	0.096
Field Efficiency, %	ND ^a	6.55
Traveling Speed, kph	ND ^a	6.94
Fuel Consumption Rate, L/h	ND ^a	6.91

^a Not specified by the manufacturer.

Note: ND – No Data

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Not provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Not provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly, and installation	Not provided
h. Specifications	Provided
i. Dismantling and disposal	Not provided
j. Warranty	Not provided
k. Parts list	Provided
2. Previous test report of similar brand and model	None
3. Manufacturer	BANKURA PRECISION AGRICULTURAL EQUIPMENTS NH 34, Opposite Tikaula Sugar Mills Ltd., Tikaula, Uttar Pradesh, India
4. Four-wheel tractor used	37.4 kW FARMTRAC 6050 C
5. Serial number	Not provided
6. Number of operators	One
7. Stability of the tractor during land leveling	The tractor was stable during land leveling.
8. Safety features	None
9. Failures or abnormalities that may be observed on the implement or its component parts during and after the operation	None
10. Other remarks	The land leveling operation required multiple overlapping passes to complete.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Dates of Test	January 24-25, 2023
Location of Test	Luisita Access Road, Tarlac City, Tarlac 15°26'08.36" N, 120°39'12.56" E
Items Tested	Operator's Manual, Actual Field Capacity, Field Efficiency, Traveling Speed, Noise Level, Fuel Consumption, Coefficient of Variation, Quality of Work, and Number of Operators

METHODS OF TEST

The laser leveler was tested based on the AMTEC Methods of Test.

DESCRIPTION OF THE IMPLEMENT

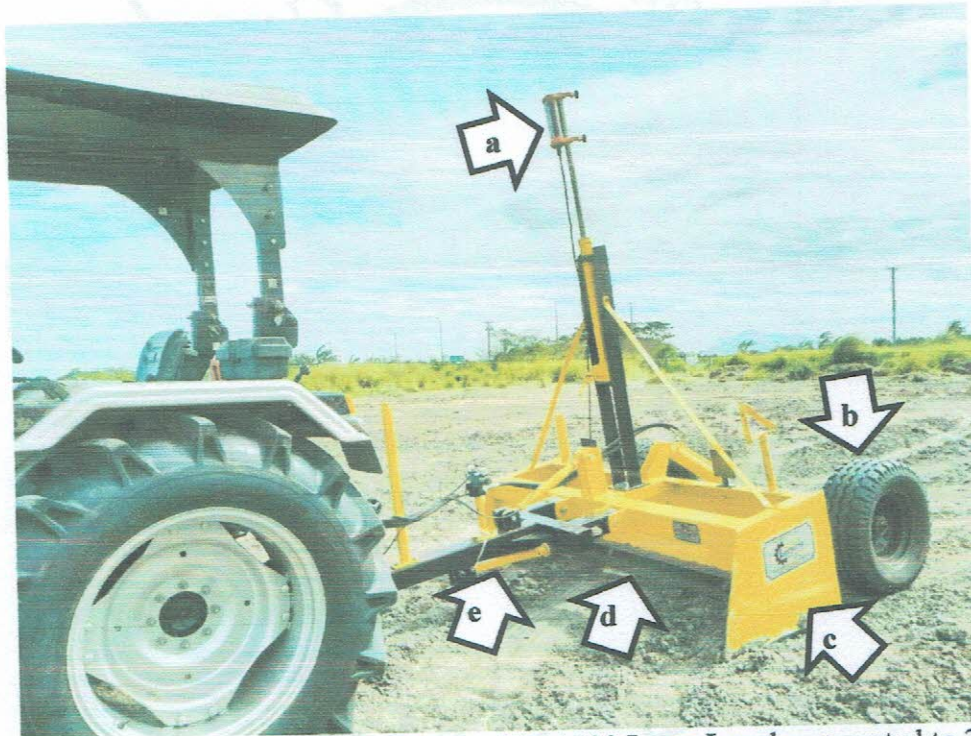


Figure 1. Parts of the LANDKING ALPHA XD-1000 Laser Leveler mounted to 37.4 kW FARMTRAC 6050 C include the following: (a) Laser receiver, (b) Transport wheel, (c) Bucket/Scraper, (d) Scraping blade, and (e) Stand.

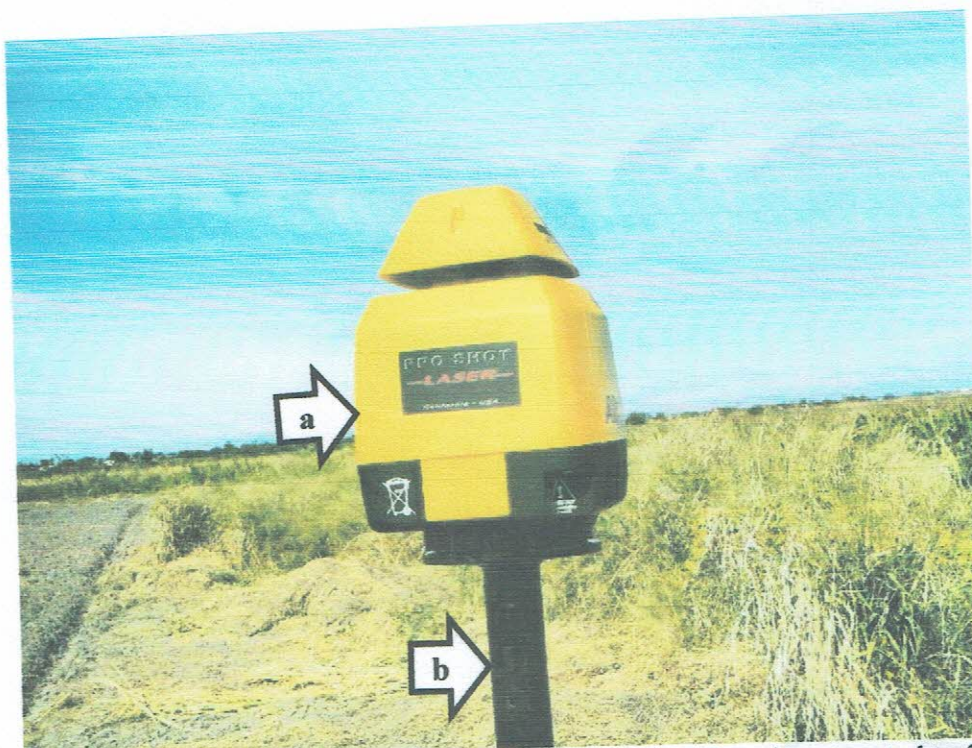
DESCRIPTION OF THE IMPLEMENT (Cont'n)

Figure 2. Parts of the LANDKING ALPHA XD-1000 Laser Leveler mounted to 37.4 kW FARMTRAC 6050 C include the following: (a) Laser transmitter and (b) Laser transmitter stand/holder.



Figure 3. Controller of the LANDKING ALPHA XD-1000 Laser Leveler.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Overall dimensions, mm		
1.1	Length	ND	3360
1.2	Width	ND	2125
1.3	Height	ND	2878
2	Weight, kg	ND	NM
3	Drag bucket		
3.1	Dimensions, mm		
3.1.1	Length	ND	770
3.1.2	Width	ND	2120
3.1.3	Height	ND	609
3.1.4	Sheet thickness	ND	8.40
3.2	Bucket capacity, kg	ND	NM
4	Scraping blade		
4.1	Dimensions, mm		
4.1.1	Length	ND	125
4.1.2	Width	ND	2120
4.1.3	Thickness	ND	12.50
5	Main frame		
5.1	Dimensions, mm		
5.1.1	Square tube, L × W, mm	ND	2125 × 132
5.1.2	Frame height	ND	132
6	Laser transmitter		
6.1	Dimensions, mm		
6.1.1	Length	ND	186
6.1.2	Width	ND	129
6.1.3	Height	ND	188

^aThe requesting party did not provide complete machine specifications.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification	AMTEC Verification
7	Laser receiver		
7.1	Brand	AGRICONTROL	AGRICONTROL
7.2	Model	RC-05B	RC-05B
7.3	Dimensions, mm		
7.3.1	Length	ND	160
7.3.2	Width	ND	195
7.3.3	Height	ND	188
7.4	Type	ND	Electronic
8	Controller		
8.1	Brand	AGRICONTROL	AGRICONTROL
8.2	Model	C5	C5
8.3	Dimensions, mm		
8.3.1	Length	ND	120
8.3.2	Width	ND	160
8.3.3	Height	ND	45
8.4	Control option	Manual or Automatic control	With automatic and manual control for raising and lowering the bucket.
9	Hydraulic control system	ND	Embedded to controller
10	Tires		
10.1	Number	2	2
10.2	Size	340/50-16	340/50-16
11	Leveling width, mm	ND	2120
12	Safety devices	ND	None
13	Special features	ND	None

*The data were obtained from the sticker of the machine.

Note: ND – No Data

RESULTS

Item		Data	
1	Test conditions		
1.1	Location of test	Luisita Access Road, Tarlac City, Tarlac 15°26'08.36" N, 120°39'12.56" E	
1.2	Type of soil	Sandy loam	
1.3	Field dimensions, L × W, m	46 × 23	
1.4	Area, m ²	1058	
1.5	Soil resistivity, kg/cm ²	NA	
1.6	Weed density	NA	
2	Field performance		
2.1	Tractor gear setting	4 th gear – Low	
2.2	Traveling speed, kph	6.94	
2.3	Type of operation	Land leveling	
2.4	Effective width of cut, mm	2104	
2.5	Actual field capacity		
2.5.1	ha/h	0.096	
2.5.2	h/ha	10.46	
2.6	Theoretical field capacity		
2.6.1	ha/h	1.46	
2.6.2	h/ha	0.685	
2.7	Field efficiency, %	6.55	
2.8	Noise level, dB(A)	91.5	
2.9	Fuel consumption		
2.9.1	L/h	6.91	
2.9.2	L/ha	99.10	
2.10	Field operational pattern	Circuitous pattern rounded corners	
2.11	Coefficient of variation, %	Trial 1	Trial 2
2.11.1	Before leveling	8.44	12.93
2.11.2	After leveling	6.65	8.85
2.12	Number of rounds	54.0	155.5

Note: NA – Not Applicable

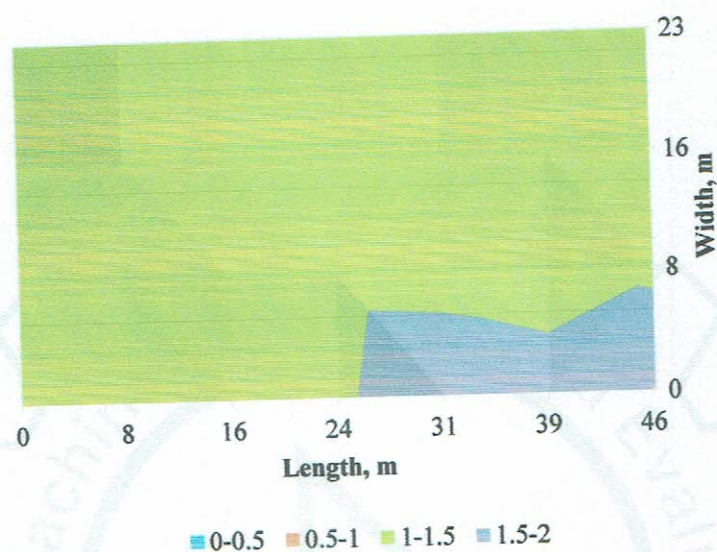
RESULTS (Cont'n)

Figure 4. Elevation of field before leveling for trial 1.

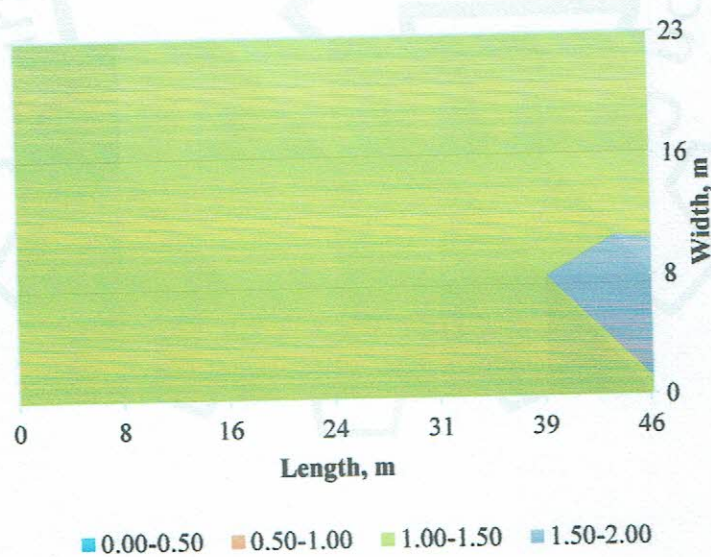


Figure 5. Elevation of field after leveling for trial 1.

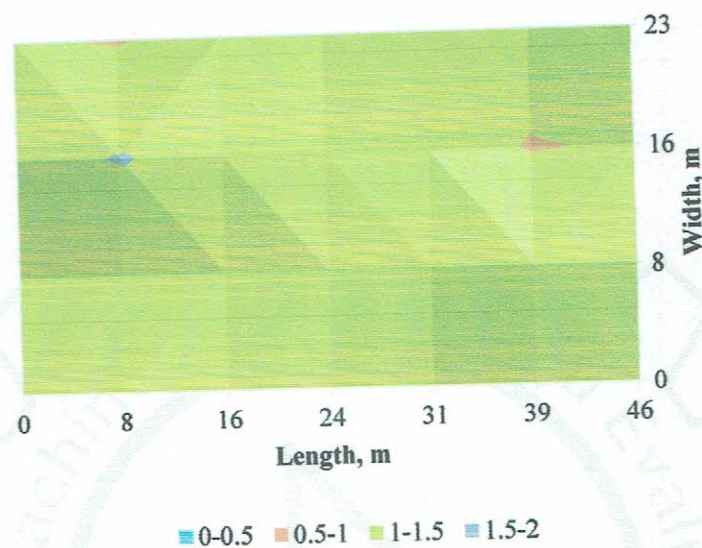
RESULTS (Cont'n)

Figure 6. Elevation of field before leveling for trial 2.

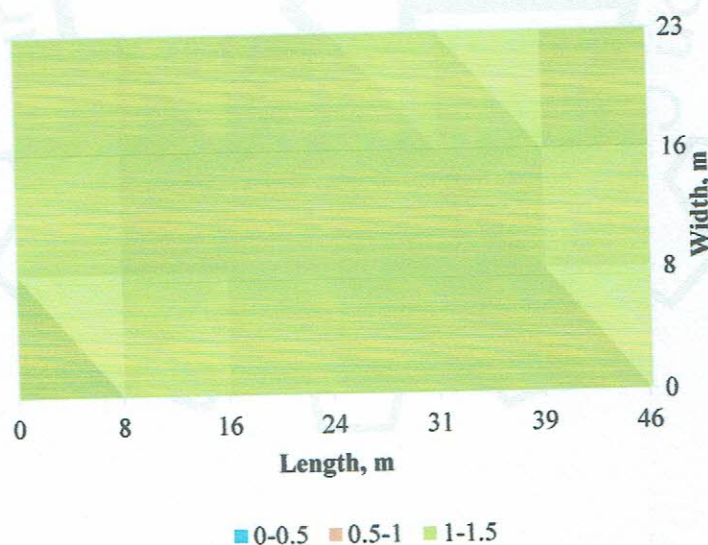


Figure 7. Elevation of field after leveling for trial 2.

OBSERVATIONS



Figure 8. Soil condition before leveling.



Figure 9. Quality of work.

OBSERVATIONS (Cont'n)

Figure 10. Nameplate of the leveler.



Figure 11. Nameplate of the laser transmitter.

OBSERVATIONS (Cont'n)

Figure 12. Nameplate of laser receiver.



Figure 13. Nameplate of the control unit.

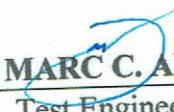
OBSERVATIONS (Cont'n)

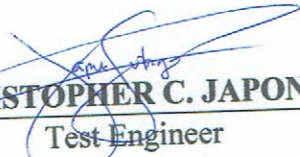
Figure 13. Warning sticker on the leveler.



Figure 14. Marking and warning sticker on the leveler.

Tested by:

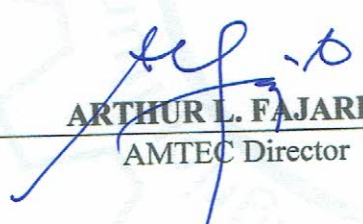

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FEB 22 2023
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